FinRelief AI

Automated Debt Diagnostics & Creditor Settlement Advisory

Capstone Project Review & Technical Demonstration

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Institution: Vishnu Institute of Technology, Bhimavaram
Architecture: Python (FastAPI) • SQLite (SQLAlchemy) • Google Gemini 2.5 Flash • TOTP 2FA

Distressed Borrowers Lack Objective, Affordable Debt Advisory

Core Consumer Hurdles in India & Target Demographic

Core Consumer Challenges

- Private counseling firms charge 15%–25% upfront fees.
- Borrowers lack transparent legal benchmarks and realistic settlement targets.
- Negative cash flow drives severe psychological anxiety and debt compounding.
- Predatory collection agencies exploit consumer ignorance of legal rights.

Target User Profile (India)

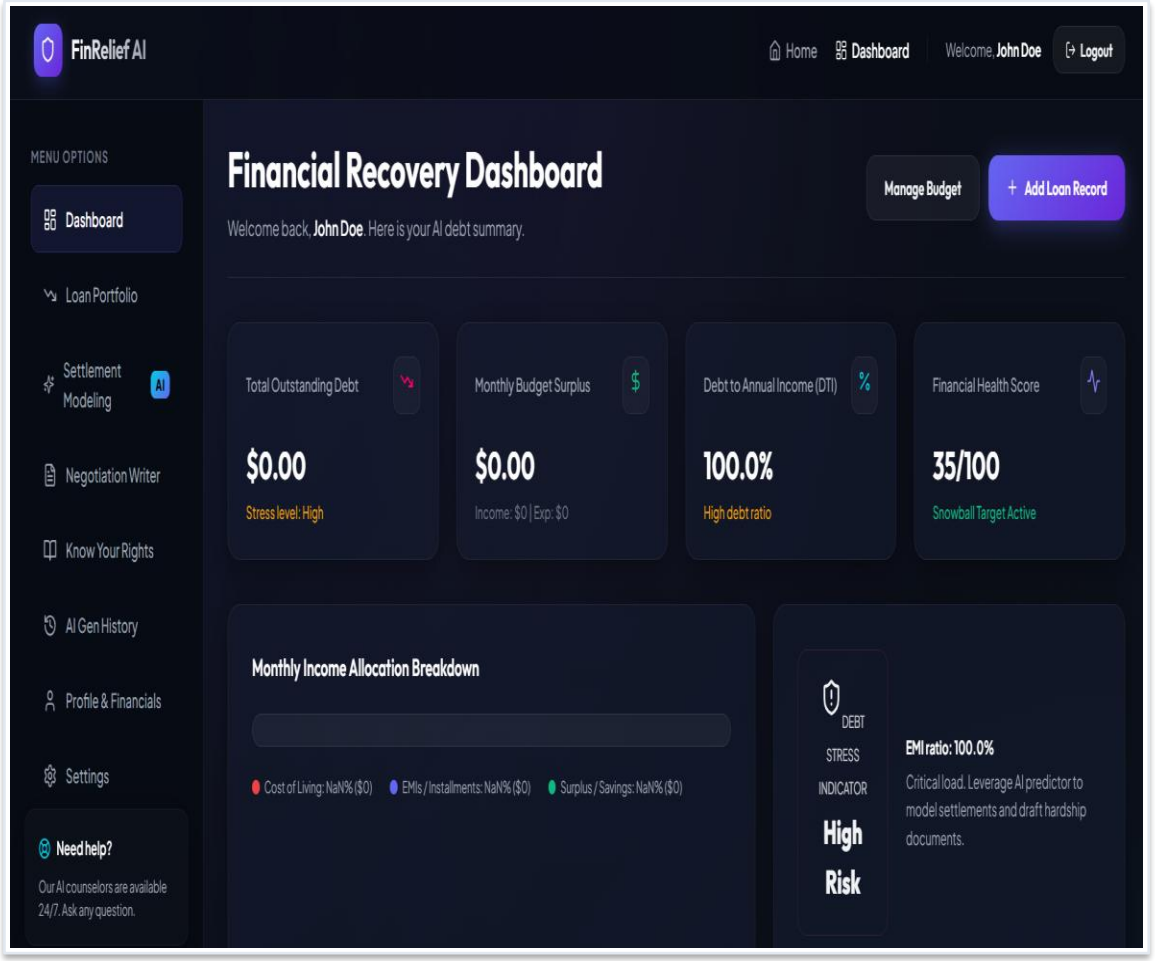
- Salaried and self-employed individuals with multiple unsecured credit accounts.
- Monthly EMI commitments exceeding 40% of gross monthly earnings.
- Accounts 30 to 180+ days delinquent seeking structured settlement.
- Borrowers seeking private, structured recovery without expensive intermediaries.

Autonomous Self-Service Web Platform for Debt Diagnostics

Working Prototype: Data Ingestion to Actionable Settlement Roadmap

Three Core Capabilities

- Instant Health Scoring:
  Real-time computation of cash surplus, DTI, and 1–100 health score.
- Heuristic Settlement Targets:
  Calculates realistic ~40%–50% lump-sum benchmarks for negotiations.
- Automated Hardship Letters:
  Generates structured communication drafts tailored by tone for creditors.
- Educational FDCPA Library:
  Direct legal rights references protecting borrowers from harassment.



Decoupled Architecture with High-Availability AI Fallback

FastAPI Asynchronous Gateway & Resilient Multi-Tier Infrastructure

Client Presentation Layer

- Single-page application (HTML5/CSS/JS).
- Responsive glassmorphism design system.
- Bearer JWT auth header injection.
- Interactive Chart.js visualizations.

FastAPI Application Core

- Asynchronous Python 3.10 REST gateway.
- Strict Pydantic v2 schema validation.
- OAuth2 Password flow & PyOTP 2FA.
- SlowAPI endpoint rate limiting.

Dual-Engine AI Coordinator

- Primary: Google Gemini 2.5 Flash SDK.
- Automated circuit-breaker monitor.
- Fallback: Deterministic local rule engine.
- Instant failover in under 0.1 ms.

Persistence Architecture & Decoupled State

- SQLite Database: Managed via SQLAlchemy ORM with 6 relational models (Users, Loans, AIHistory, Profiles, Settlements, Negotiations).
- Concurrency Hardening: Configured PRAGMA journal_mode=DELETE, synchronous=NORMAL, and 30s connection timeout to eliminate locks.
- Modular Decoupling: Deterministic math runs entirely in Python prior to any external AI invocation.

Deterministic Algebraic Logic Outperforms LLM Guessing

Clean Mathematical Models for Real-Time Financial Diagnostics

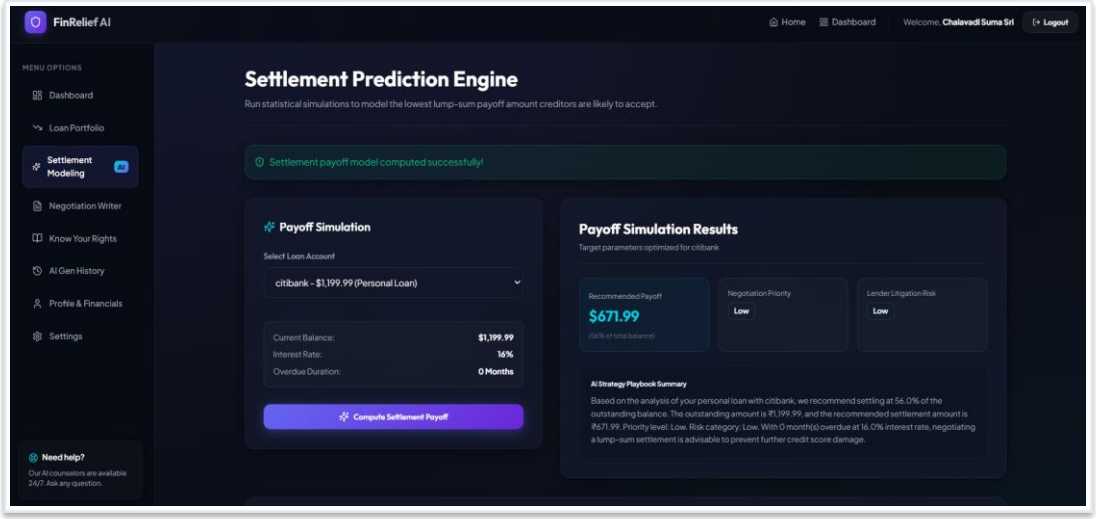
Core Formulas & Scoring Algorithm

Disposable Monthly Surplus:
Surplus = Income - (Living Expenses + Total EMIs)

Debt-to-Income (DTI) Ratio:
DTI = (Total Monthly Loan EMIs / Gross Income) * 100

Financial Health Score (1–100):
Base(100) penalized by high DTI and delinquency; credited by positive cash surplus.

Stress Categorization:
Classified into Low, Medium, High, or Critical Stress.



Worked Example (Indian Rupees ₹)

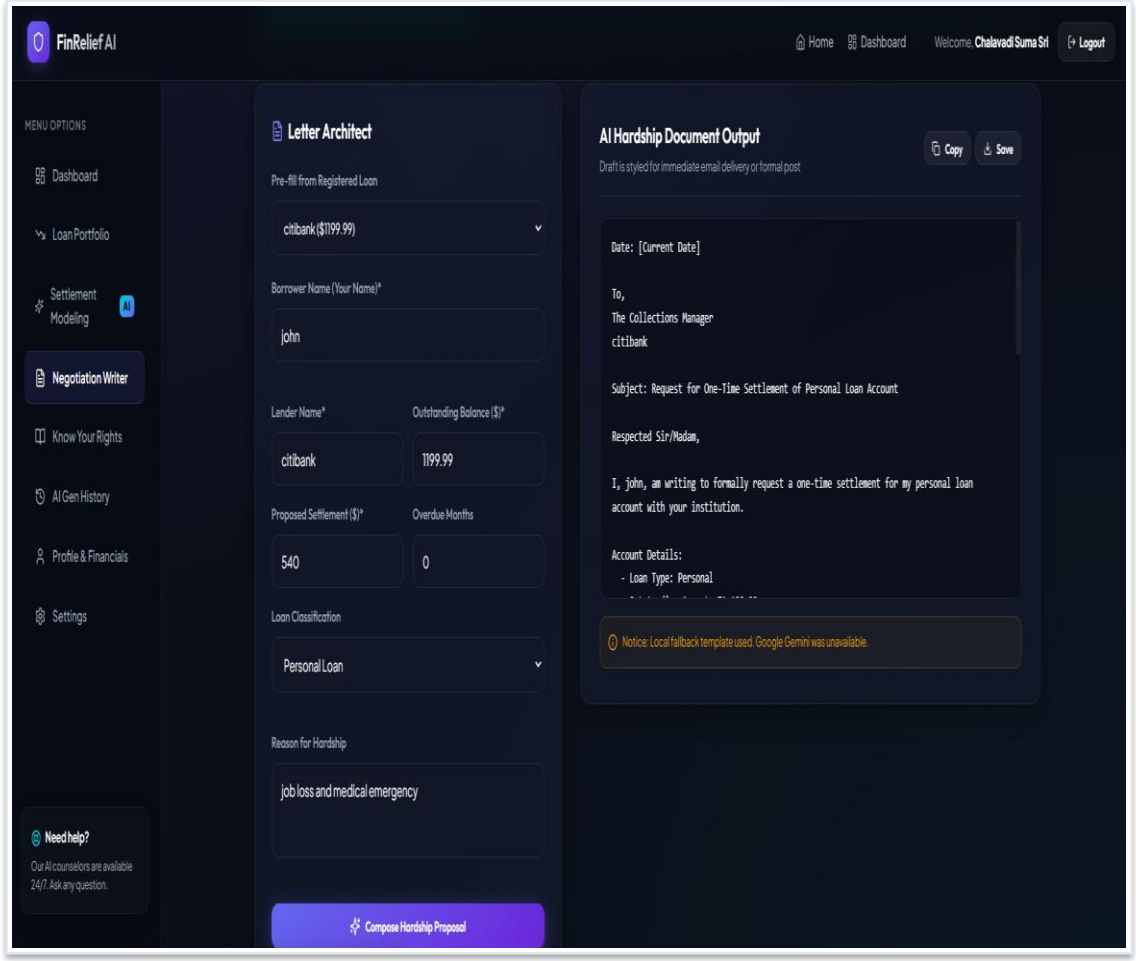
- Monthly Income: ₹50,000 | Living Expenses: ₹25,000 | EMIs: ₹15,000
- Calculated Surplus: +₹10,000 | DTI: 30.0% (Moderate)
- Health Score: 80 / 100 (Low Stress Profile)
- Critical Case: If EMIs reach ₹30,000 -> Surplus -₹5,000 -> Score 20/100

AI Automatically Fails Over to Local Logic in Milliseconds

Dual-Engine Strategy Guarantees Continuous Letter Generation

Dual-Engine Workflow

- Primary Engine Call:
  Dispatches prompt to Google Gemini 2.5 Flash with tone control (Professional/Firm/Conciliatory).
- Automated Circuit Breaker:
  Catches APIError, 429 quota exhaustion, or timeouts in under 0.1 ms.
- Local Rule-Based Fallback:
  Generates complete structured hardship letter using verified deterministic templates.
- Audit & Disclaimer:
  Logs letter to ai_history table; explicitly disclaims letter as negotiation draft, not legal advice.



Defense-in-Depth Security Aligned with OWASP Standards

Multi-Tier Protection Safeguarding Sensitive Borrower Records

Signed Token Authentication

- Stateless session management via HS256-signed JWTs.
- 60-minute token expiration enforced server-side.
- Stored in client localStorage (HttpOnly cookie migration planned).
- Strict Bearer token header verification on protected endpoints.

Two-Factor Auth (RFC 6238 TOTP)

- Google Authenticator integration via pyotp library.
- Base32 secret generation with QR code provisioning.
- Enforces time-sensitive 6-digit one-time codes on login.
- Step-up intermediate tokens prevent unauthorized access.

Bcrypt Password Hashing

- Direct bcrypt hashing with dynamic salt rounds.
- ~197 ms measured execution work factor prevents brute-force.
- Completely eliminated legacy unmaintained passlib wrapper.
- Plaintext passwords never written to database or logs.

Abuse Control & CORS Policy

- SlowAPI rate-limiting (3/min on signup, 5/min on login).
- Explicit CORS origin whitelisting for trusted client domains.
- Pydantic v2 input sanitization rejects negative financial values.
- Local SQLite database prevents third-party data telemetry.

Automated Test Suite Confirms 100% Pass Across Core Routes

Empirical Verification Measured Locally with Pytest 8.2.2 & Mocked Fixtures

6 / 6 Passed	3.48 s	3.8 ms	< 0.1 ms
Pytest Test Suite	Test Suite Runtime	Avg API Health Latency	AI Failover Execution

Test Function Name	Endpoint / Component	Scope / Assertion Tested	Status
test_health_check	GET /health	Validates server availability & healthy JSON payload	PASSED
test_signup_and_login	POST /api/signup & /login	Bcrypt password verification & JWT token issuance	PASSED
test_create_and_get_loan	POST & GET /api/loans	Authenticated loan creation and user loan array query	PASSED
test_settlement_recommendation	POST /api/loans/{id}/settlement	Calculates EMI ratio, heuristic target & narrative	PASSED
test_negotiation_letter	POST /api/loans/{id}/letter	Tone selection & structured hardship letter generation	PASSED
test_unauthorized_access	GET /api/loans (No JWT)	Enforces strict HTTP 401 Unauthorized rejection	PASSED

Overcoming Critical Concurrency, Dependency & State Hurdles

Pragmatic Engineering Solutions Implemented During System Integration

1. SQLite Concurrency in Cloud-Synced Folders

- Problem: Running SQLite inside cloud-synced folders (OneDrive) locked WAL temporary files, causing 'database is locked' errors.
- Solution: Switched PRAGMA journal_mode=DELETE, synchronous=NORMAL, and set a 30s connection timeout to eliminate locks completely.

2. Passlib and Bcrypt Version Incompatibility

- Problem: The legacy passlib package (unmaintained since 2020) crashed with TypeError on modern bcrypt >= 4.1 during password hashing.
- Solution: Excised passlib entirely. Re-implemented password hashing directly via bcrypt.hashpw() and bcrypt.checkpw() with dynamic salt generation.

3. Dual-Engine Resilience & Rate Limiting

- Problem: Free-tier Gemini API calls risked HTTP 429 quota exhaustion or network dropouts, which would break letter generation.
- Solution: Built an automated try/except circuit breaker that diverts to local deterministic templates in 0.01 ms, ensuring continuous availability.

Working Prototype with a Defined Path to Enterprise Scale

Transparent Scope Boundaries, Scalability Roadmap, and Discussion

Current Prototype Limitations

- Single-Node SQLite:
  SQLite is single-writer; production concurrency requires PostgreSQL.
- Manual Data Ingestion:
  Users enter financial data manually; lacks live bank API synchronization.
- Client-Side Token Storage:
  JWT currently in localStorage; requires HttpOnly cookie migration.
- Heuristic Predictions:
  Settlement percentages (~40-50%) are heuristic guidelines, not legal advice.

Next Technical Milestones

- Account Aggregator (AA) Integration:
  Connect with RBI-licensed Account Aggregator APIs for automated bank statement ingestion in India.
- PostgreSQL Migration:
  Transition database to scalable PostgreSQL with connection pooling.
- CIBIL Credit Score Simulator:
  Simulate credit score recovery trajectories post-settlement.
- Snowball vs. Avalanche Planner:
  Interactive visual planner comparing payoff timelines and interest savings.

Questions & Technical Discussion — Thank You!

GitHub Repository: https://github.com/syedjafarmoinuddin786/FinRelief